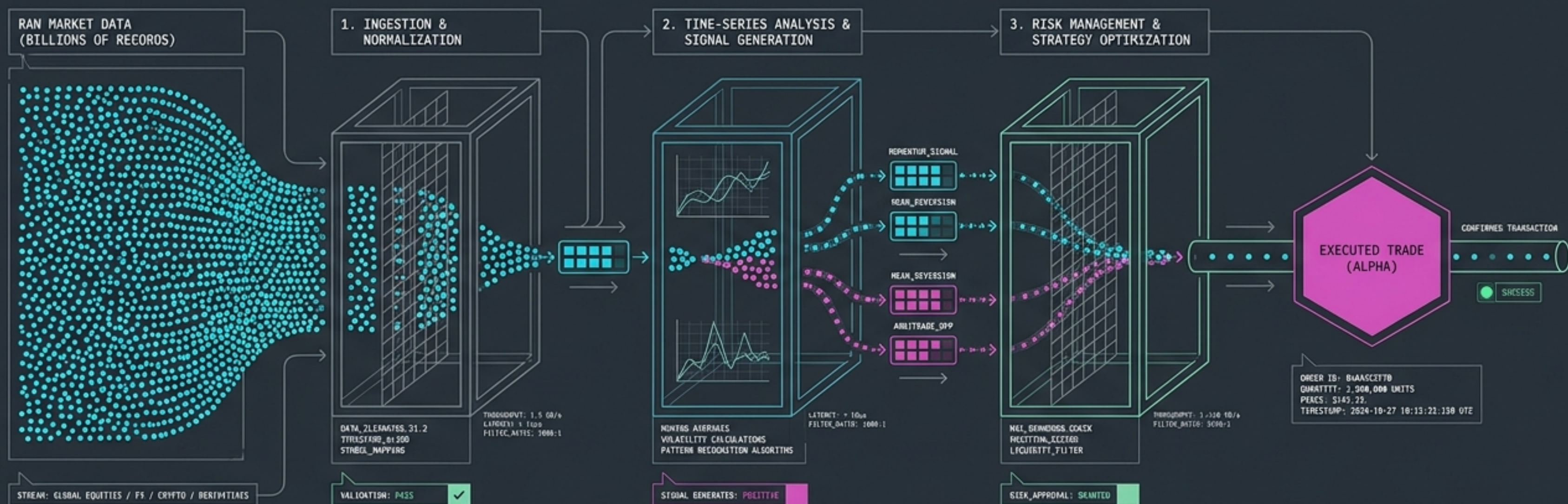
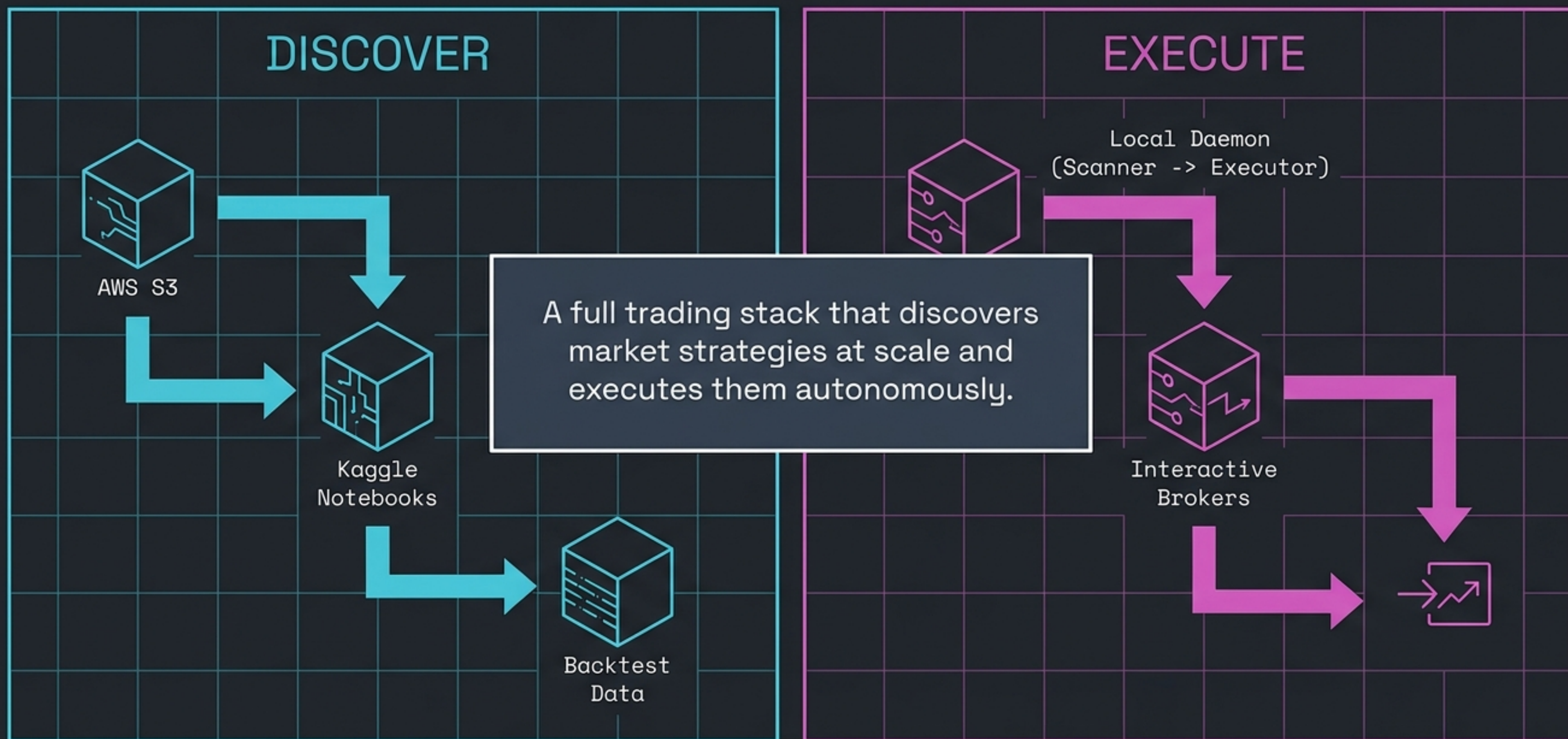


THE MARKET DATA PLATFORM

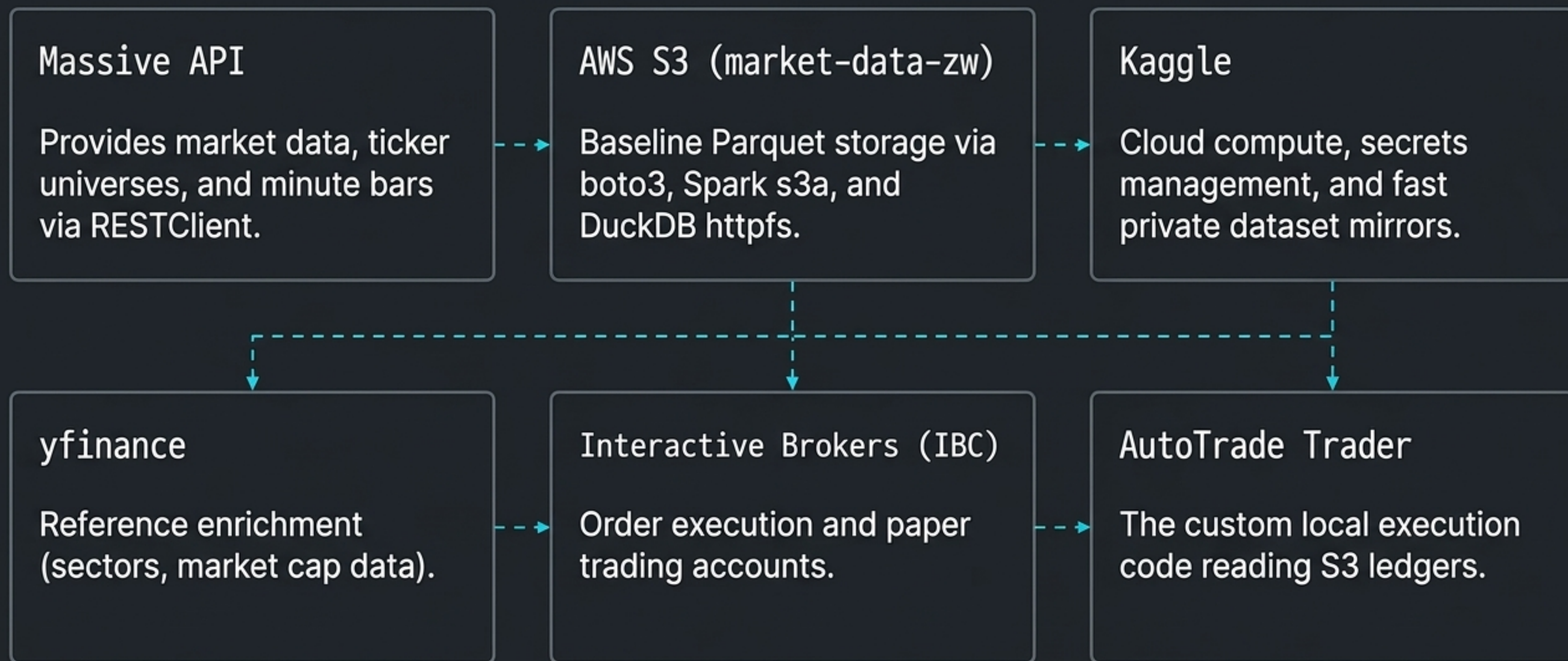
An End-to-End Quantitative Trading Stack:
From Billions of Records to Executed Alpha.



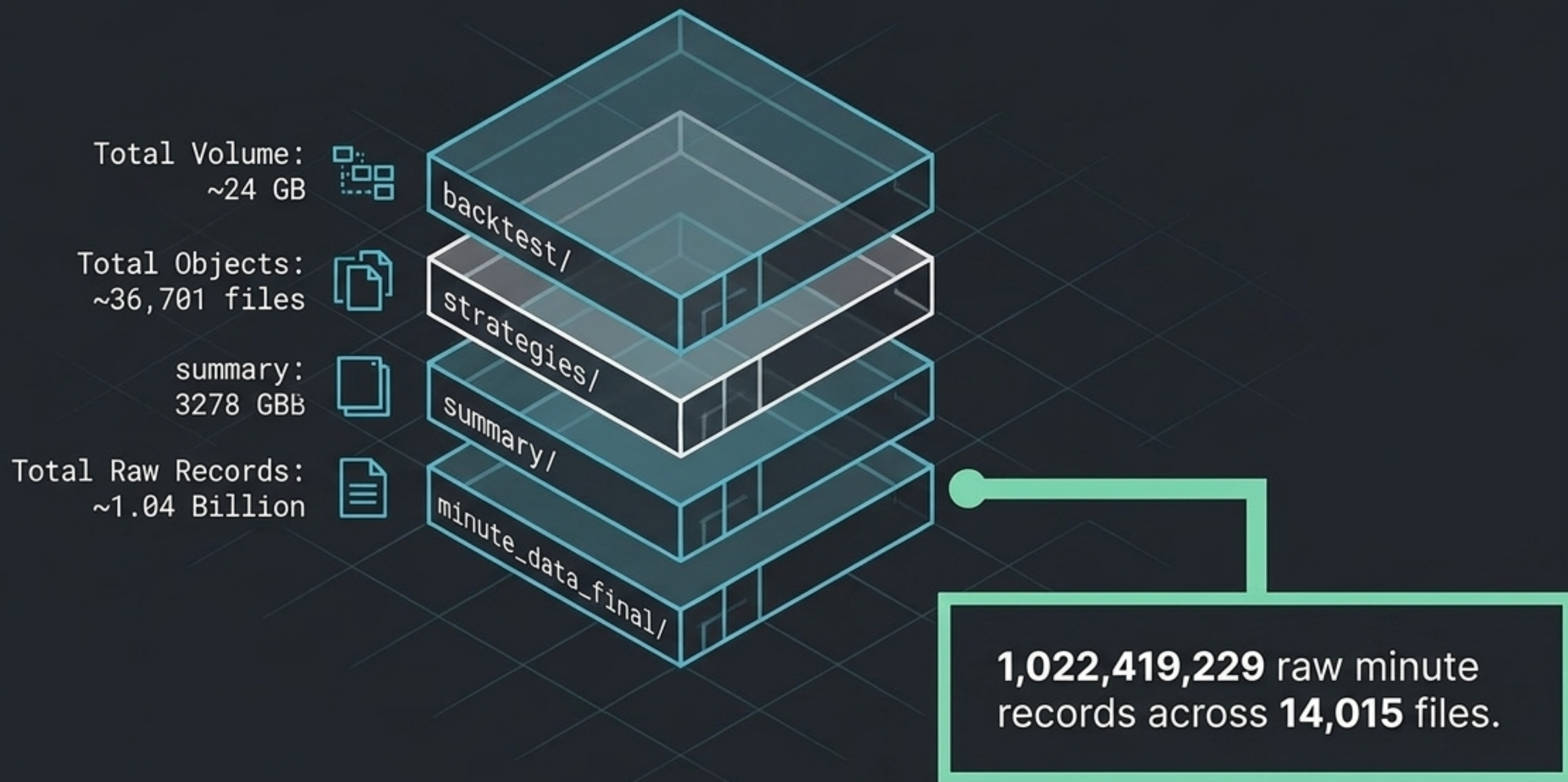
Two distinct engines powering a single trading lifecycle



The foundational tech stack and external integrations

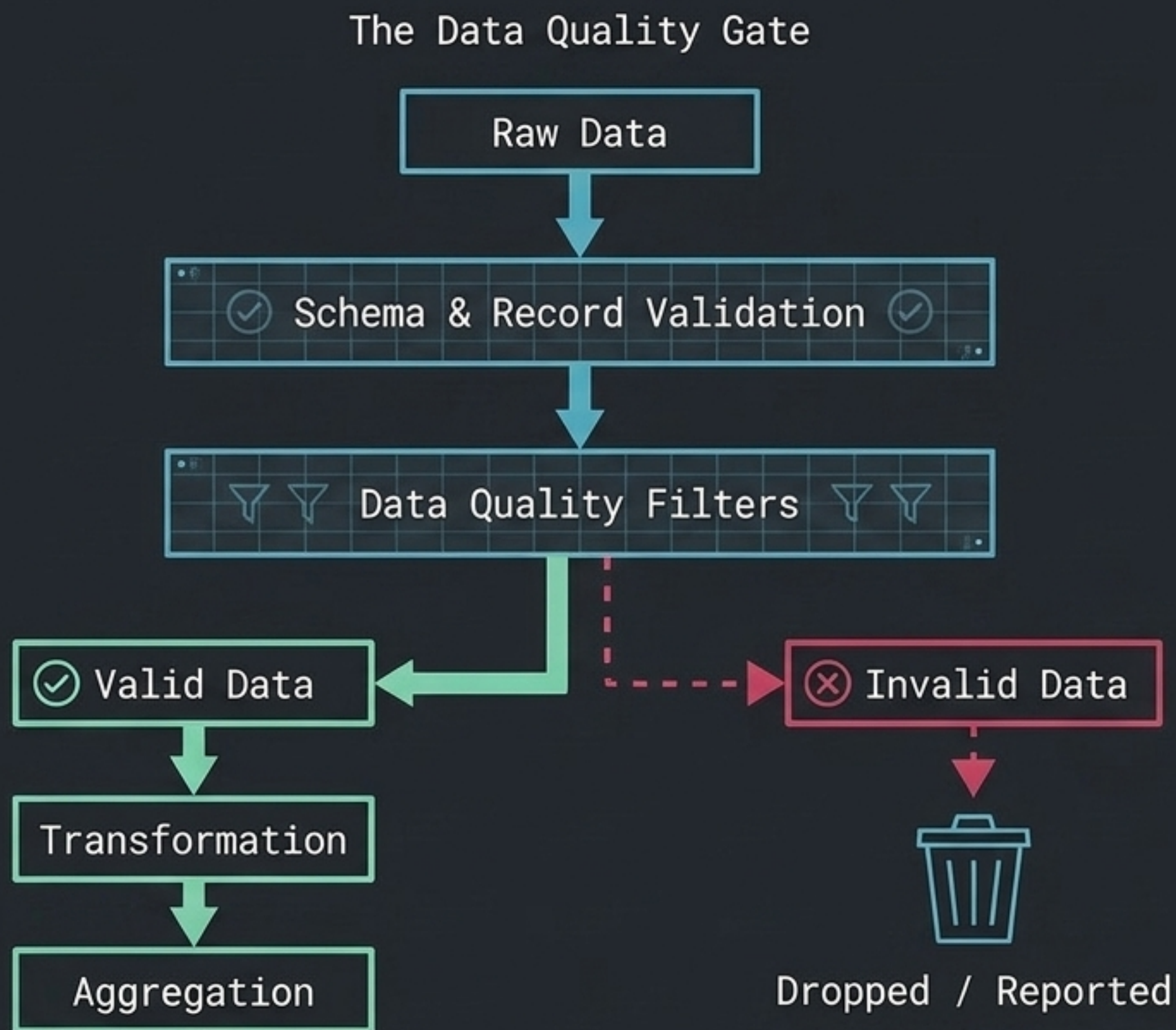


Engineering for genuine scale over fabricated estimates



Not estimates. Measured directly from actual Parquet metadata.

Fast pipelines are useless if they produce incorrect data

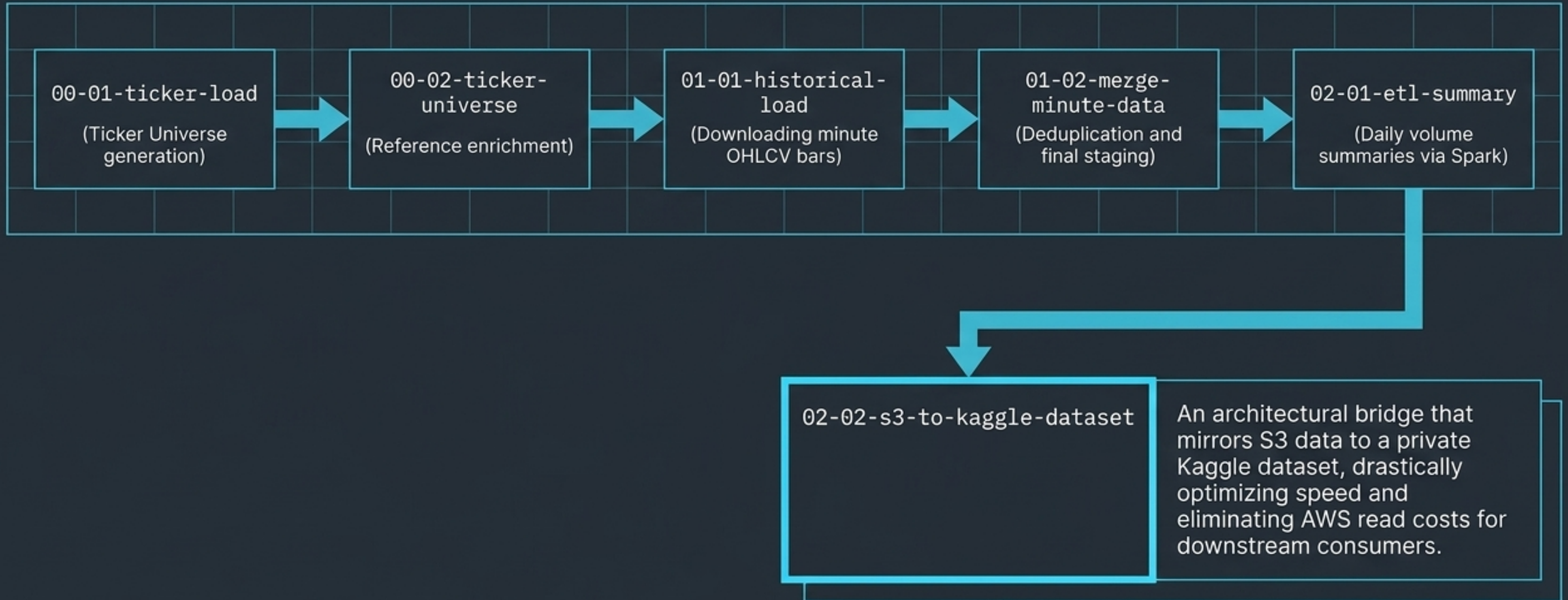


Processing Engines: Distributed scale versus local vectorization

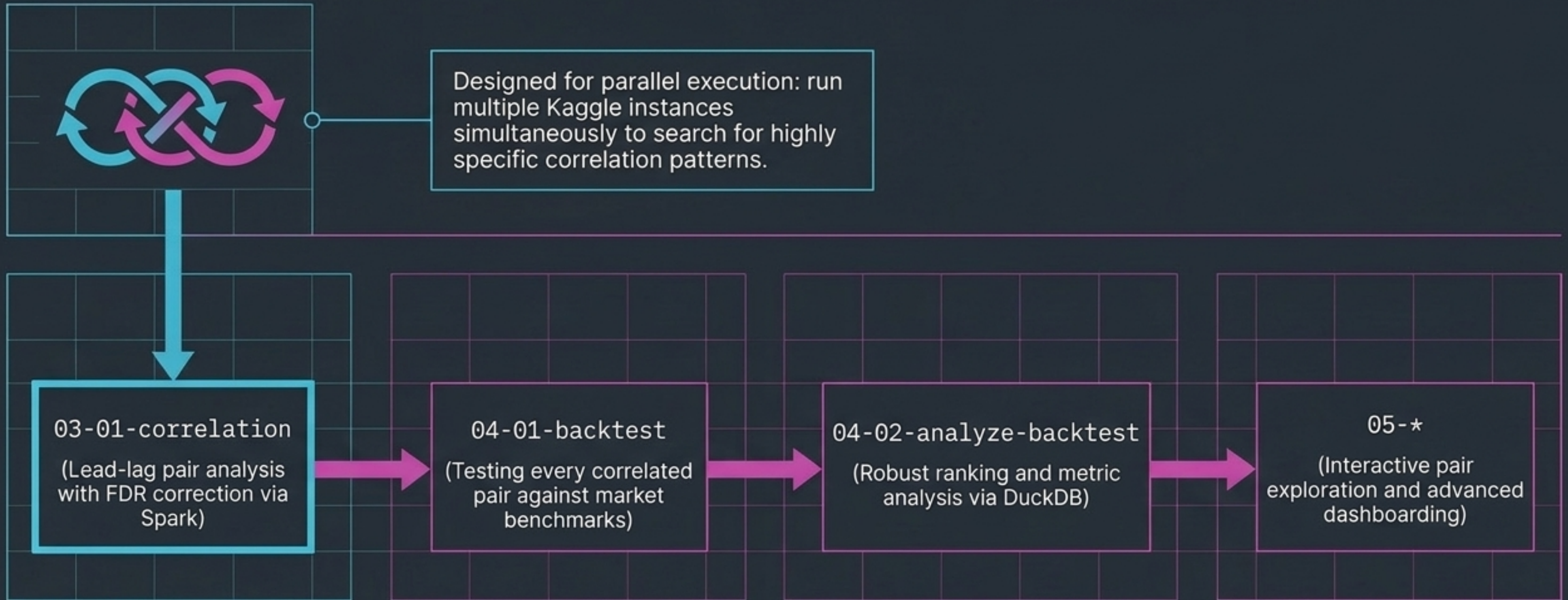
Apache Spark Distributed	DuckDB Local Vectorized
→ Use Case: Stages 02-01, 03-01, 04-01 → Mechanics: Partition pruning, predicate pushdown, shuffles, parallel aggregation. • Storage Link: Reads S3 via s3a://	→ Use Case: Stages 04-02, 05-01, 05-02 (Metrics & Exploration) → Mechanics: SQL over Parquet, single-node vectorized processing. • Storage Link: Reads S3 via httpfs.

Implemented side-by-side so the exact performance trade-offs between local execution and distributed scaling are explicitly measurable.

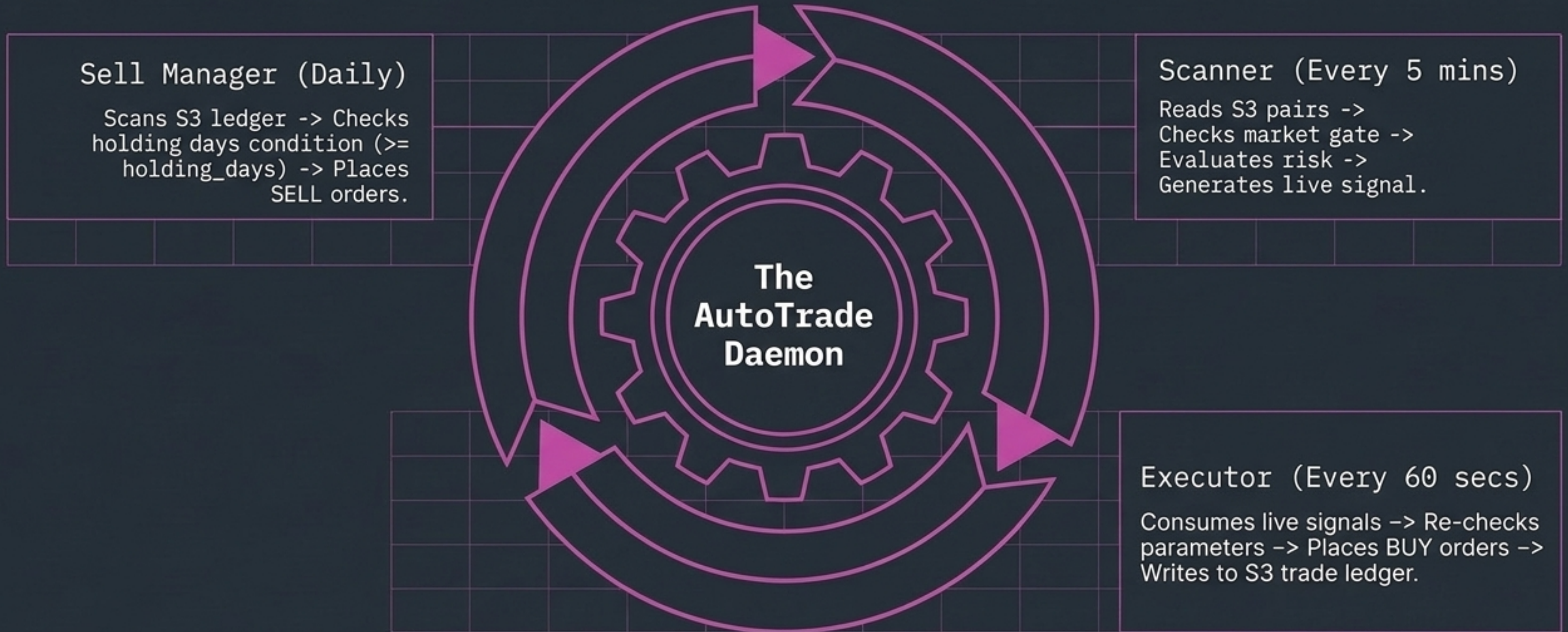
The Discover Pipeline Phase 1: Ingestion and ETL



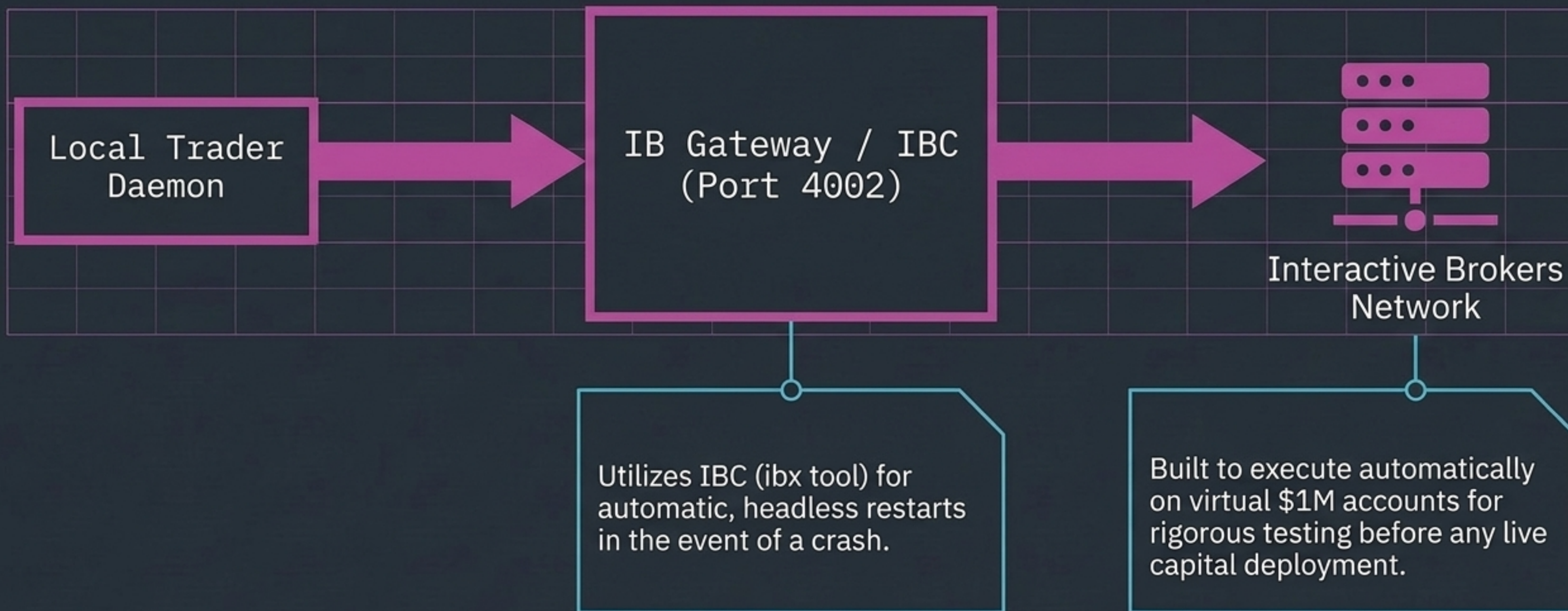
The Discover Pipeline Phase 2: Alpha Generation



The Execute Loop: Autonomous local trading architecture



Seamless market execution via Interactive Brokers

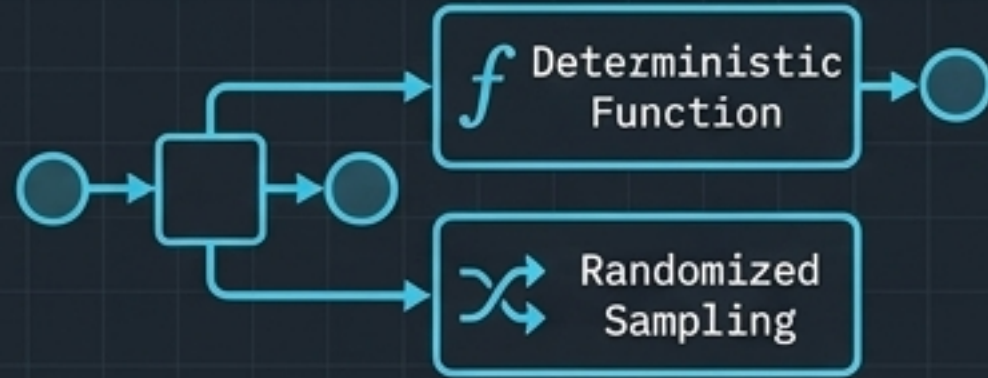


Minimal deployment overhead and strict security controls

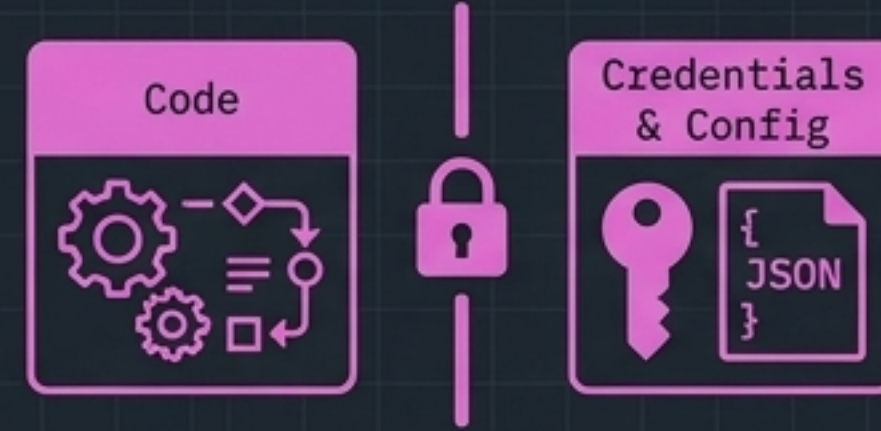
Compute	Kaggle Account	Free
Storage	AWS Account + S3 Bucket (~24GB footprint)	~\$
Data	Massive API Key	Free tier ~5 calls/min
Enrichment	yfinance	Free / Open Source
Execution	IBKR + Gateway	Paper Trading Free

Zero hard-coded secrets. All credentials are injected via environment variables (AWS_ACCESS_KEY_ID, MASSIVE_API_KEY) or isolated config.json configurations.

The four engineering principles underpinning the stack



Deterministic pipelines ensure the exact same input always yields the exact same output. Randomized sampling is strictly isolated.



Absolute separation of code and credentials.

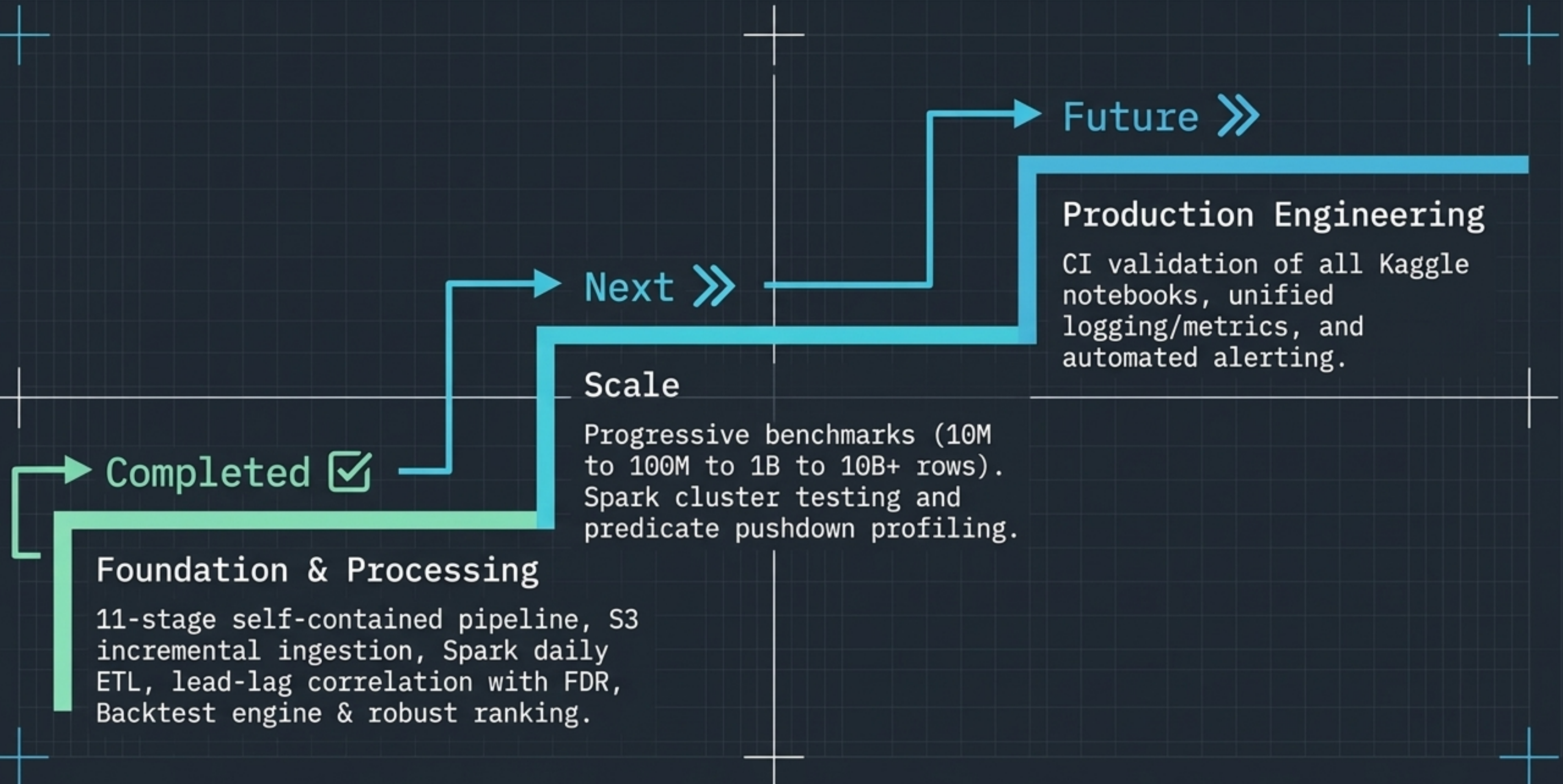


All architecture decisions—like the Spark vs. DuckDB split—are backed by measurable benchmark results.

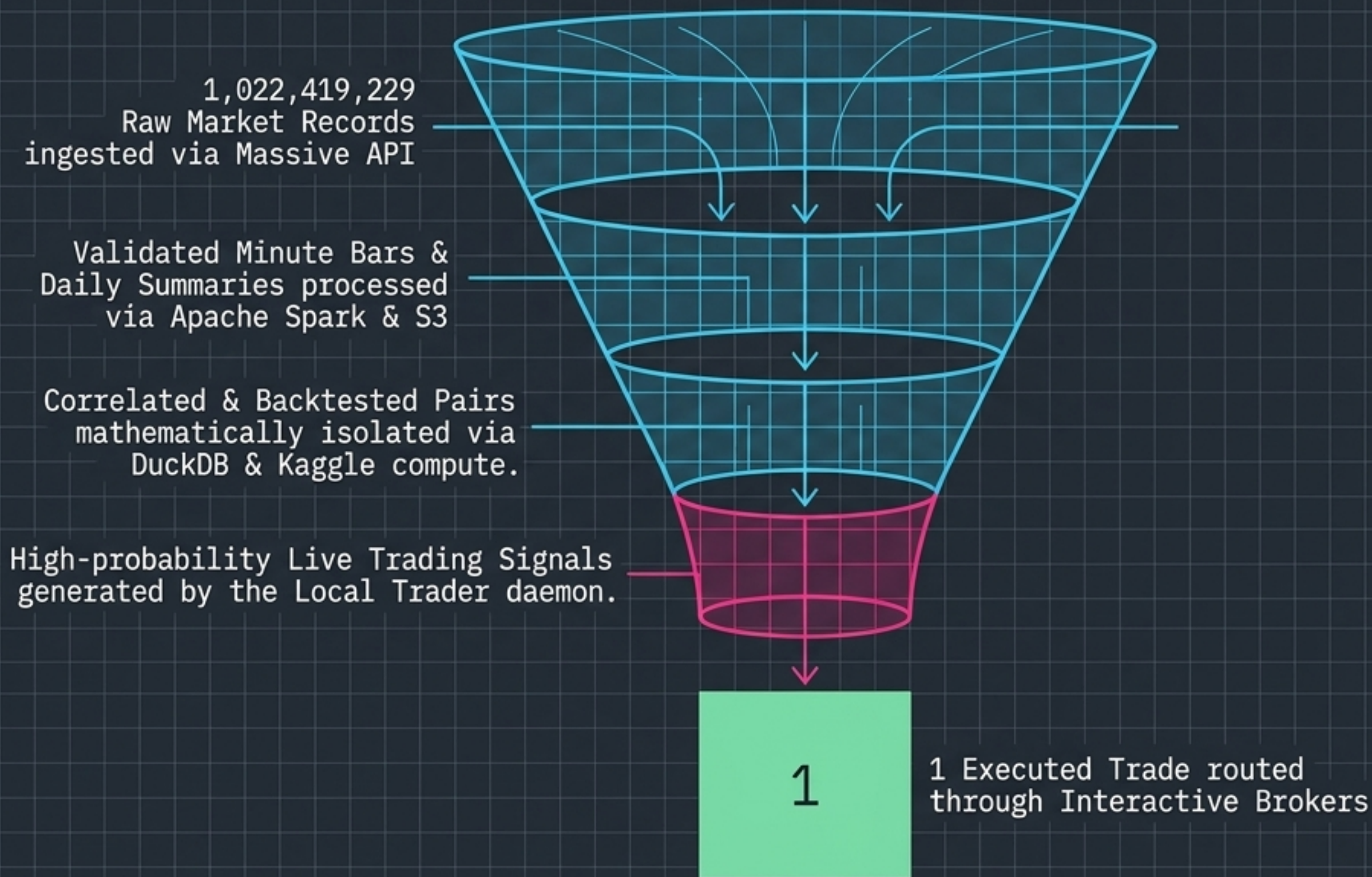


Pipelines resume intelligently from S3 object keys, aggressively avoiding redundant scans and expensive shuffles.

Platform Roadmap: From foundation to extreme scale



The Alpha Funnel: From infinite noise to executed logic



A deterministic, end-to-end engine transforming a 1,000,000,000 rows of market chaos into verifiable, automated execution